Justin Bishay

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Skills

Programming Languages: C++, C, C#, Python, JavaScript, Java, HTML/CSS

Game Engines: Tiger Engine (Destiny 2/Marathon), Unity, Unreal Engine, Godot

Tools/Software: Visual Studio, Git, Perforce

Work Experiences

Bungie — **Bellevue, WA**

Gameplay Engineer — **June 2023 - June 2026**

Systems level engineering on proprietary game engine technology

- Solved complex problems within a server, client, and host networking model to support PvE and PvP activities
- Investigated and developed new script debugging system so multiple disciplines could better debug and understand scripted content
- Created tooling for inspecting network traffic generated by content which revealed 3 instances of existing content that was contributing to heavier than acceptable network traffic
- Led technical development of matchmaking and activity systems and guided design and UI/UX teams for new PvEvP game mode in Destiny 2
- Made optimizations to reduce matchmaking times to be no more than 2 minutes for novel PvEvP activity

Collaborated across interdisciplinary teams

- Provided engineering feedback for other engineers on code reviews and for designers on scripting for raid, dungeon, and seasonal activity content to refine best coding and scripting practices and reduced live bug count on new live game releases by over 50%
- Mentored engineering intern and taught them how to navigate engine code and shepherded them on fixing a text chat bug with whisper privacy
- Took feedback from designers to identify areas where engine features were lacking and gathered requirements for new features to service their needs and improve workflows
- Evaluated the technical requirements necessary to support new content initiatives; ensured alignment with late-stage adjustments and modified deliverables accordingly across multiple ongoing projects without compromising quality
- Analyzed runtime data to find content or code that led to slowdown in frame rates to make certain that content shipped at an acceptable performance level for each release

Maintained an active development branch as a branch owner

- Set up and scheduled various types of build jobs to produce daily and weekly playtests and a steady cadence of content updates on local development builds
- Reviewed daily build reports to identify build breaks; examined an average of 3 build blockers per week and made or routed fixes to prevent the build from being blocked for more than a few hours of the week

Associate Gameplay Engineer **January 2022 - June 2023**

Worked closely with design teams to provide new engine features to support their content needs

- Rewrote existing teleporter game system to improve code legibility and support new functionalities including the ability to visual effects before and after teleportation
- Created new scripting functionality for designers to identify when specific players used a specific teleporter allowing for simpler content setups around player teleportation
- Modified sandbox systems to provide new methods for design to specify the objects a projectile should auto target thus allowing for more streamlined content creation practices

Engineering Intern **May 2021 - August 2021**

Made significant contributions to shipping cross-play in Destiny 2 and Marathon

- Developed the skillset to navigate a large codebase and identify the areas that are relevant to current feature work
- Quickly obtained a solid understanding of a proprietary game engine and improved self velocity working in the engine making it possible to take on more tasks than originally intended for an intern
- Identified areas where additional improvements could be made to the cross-play feature; proposed adding an unblock text chat command to the team and implemented it
- Gained and refined expertise on the text chat system and was able to share knowledge with other engineers to assist with their text chat related tasks

Laboratory for Advanced Visualization Applications **Honolulu, HI**

Graduate Research Assistant **August 2019 - December 2021**

Implemented cloud pixel streaming application for data visualization applications across several types of hardware clients for US Military research

- Coordinated project responsibilities with professionals and other students to solve problems around client performance, networking render data, and streamed imagery resolution
- Learned how to navigate the layer between C# and C++ code when networking data across processes built with different programming languages

Undergraduate Research Assistant **January 2018 - May 2019**

Built augmented reality training application requested by the US Navy

- Joined midway through project and collaborated with computer engineering student to learn the existing codebase and how to develop for augmented reality technologies
- Drove feature development of several core functionalities including audio recording and playback, recorded videos panels in 3D virtual space, and 3D object placement and editing
- Used feedback from a PhD student to overhaul UI/UX to better organize menu functions and make it smoother to use

Education

M. S. in Computer Science **December 2021, University of Hawai‘i at Mānoa**
B. S. in Computer Science **August 2019, University of Hawai‘i at Mānoa**